

CUSTOMER QUERY ANALYSIS AND NLP CHATBOT

Internship Project Report

Data Analysis • Natural Language Processing • Machine Learning

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Project	Customer Query Analysis and NLP Chatbot
Programming Language	Python
Environment	Google Colab
Dataset Size	24 customer queries

1. Project Objective

The objective of this project is to develop a simple customer-support chatbot using Python, NumPy, Natural Language Processing (NLP), TF-IDF text representation, and Logistic Regression. The project also performs data analysis and visualization on the customer-query dataset. The chatbot classifies a user query into a predefined category and generates an appropriate response.

2. Tools and Technologies Used

Tool / Library	Purpose
Python	Programming and implementation
NumPy	Numerical operations and random response selection
Pandas	Dataset creation and data analysis
NLTK	Natural Language Processing support
Matplotlib	Data visualization
Seaborn	Statistical visualization
Scikit-learn	TF-IDF, train/test split, Logistic Regression and evaluation
Google Colab	Development and execution environment

3. Dataset Description

A small dataset was created specifically for the project. It contains query, category, and response fields. There are 24 queries distributed across six categories: greeting, services, pricing, support, thanks, and goodbye. Each category contains four queries, making the dataset balanced.

Category	Number of Queries
greeting	4
services	4
pricing	4
support	4
thanks	4
goodbye	4
Total	24

4. Methodology

1. Import libraries.
2. Create the customer-query dataset using Pandas.
3. Analyze shape, columns, missing values and category distribution.
4. Preprocess text by lowercasing, removing special characters and trimming spaces.
5. Convert cleaned text into TF-IDF features.
6. Split the data into training and testing sets using a stratified 75/25 split.
7. Train a Logistic Regression classification model.
8. Predict test categories and evaluate performance.
9. Build a responding chatbot that predicts a category and selects a response.
10. Add handling for clearly unrelated questions.
11. Analyze query length and word frequency.
12. Test the chatbot with new queries.

5. Data Analysis Outputs

The dataset contains 24 records and six balanced categories. The category-count output was:

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Number of queries in each category:
greeting 4
services 4
pricing 4
support 4
thanks 4
goodbye 4
```

The category distribution graph showed equal bars for all six categories, confirming that the created dataset is balanced.

6. Query Length Analysis Output

Recorded descriptive statistics:

Statistic	Value
Count	24
Mean	3.291667 words
Standard deviation	1.545798
Minimum	1 word
25th percentile	2 words
Median	4 words
75th percentile	4 words
Maximum	6 words

Examples included: "Hello" = 1 word, "Good morning" = 2 words, "What services do you provide?" = 5 words, and "What can you help me with?" = 6 words.

7. NLP and Machine Learning

The cleaned customer queries were converted into TF-IDF vectors so the text could be processed numerically. Logistic Regression was trained to classify queries into six predefined categories. A stratified 75% training and 25% testing split produced 18 training examples and 6 testing examples.

8. Model Evaluation — Clear Output

The final recorded model result was:

Algorithm: Logistic Regression
Text representation: TF-IDF
Test Accuracy: 66.67%

Confusion matrix:

Actual \ Pred.	goodbye	greeting	pricing	services	support	thanks
goodbye	0	0	0	0	0	1
greeting	0	1	0	0	0	0
pricing	0	0	1	0	0	0
services	0	0	0	1	0	0
support	0	0	0	0	1	0
thanks	0	0	0	0	1	0

The matrix contains four correct predictions out of six test examples, giving 66.67% test accuracy. Because the test set contains only six examples, individual predictions have a large effect on the percentage.

9. Chatbot Response Testing — Actual Outputs

The completed chatbot was tested with representative customer queries. The following outputs were observed:

User: Hello

Chatbot: Hello! What can I do for you?

User: How much is the price?

Chatbot: Our charges depend on the selected service.

User: I need some help

Chatbot: I'm sorry you are facing a problem. Please provide more details.

User: Thank you

Chatbot: Glad I could help!

User: What is the weather on Mars?

Chatbot: Sorry, I can only answer questions about our services, pricing, support, and general customer queries.

10. New-Query Testing

New wording was also tested. The recorded new-query test accuracy was 66.67%. Two examples were not classified as expected: “I am facing an issue” and “See you later”. This demonstrates the limitation of having only four examples per category.

New Query	Expected	Observed
Hi there	greeting	greeting
Can you tell me about your services?	services	services
How much will I have to pay?	pricing	pricing
I am facing an issue	support	not recognized correctly
Thanks for helping me	thanks	thanks
See you later	goodbye	not recognized correctly

11. Interactive Chatbot

An interactive loop was implemented so the user can enter multiple messages. The conversation continues until the user enters “bye”. The chatbot predicts a category and returns a suitable response.

12. Handling Unrelated Questions

A simple out-of-scope check was added for clearly unrelated topics. For example, “What is the weather on Mars?” was rejected with an appropriate out-of-scope message instead of receiving a customer-service response.

13. Final Results Summary

Item	Final Result
Total queries	24
Categories	6
Average query length	3.29 words
Minimum query length	1 word
Maximum query length	6 words
Text representation	TF-IDF
Machine learning algorithm	Logistic Regression
Test accuracy	66.67%
New-query test accuracy	66.67%

Chatbot	Working interactive chatbot
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14. Limitations

- The dataset contains only 24 queries.
- Each category has only four examples, limiting generalization.
- The six-example test set makes accuracy sensitive to individual predictions.
- The chatbot supports only predefined customer-service categories.
- This is an educational prototype, not a production customer-support system.

15. Conclusion

The project successfully developed a simple customer-query classification and responding chatbot using Python, NLP, TF-IDF and Logistic Regression. It also demonstrated data analysis through category distribution, query-length analysis and word-frequency analysis, together with model evaluation using accuracy and a confusion matrix. The final chatbot can respond to supported customer queries, operate interactively, and reject clearly unrelated questions. The 66.67% evaluation results should be interpreted in the context of the intentionally small 24-query educational dataset.

16. Final Submission Contents

Recommended submission package:

1. Google Colab notebook (.ipynb)
2. Project Report (PDF)
3. Dataset/code files used by the project
4. GitHub repository with README